EX.NO: 4

DATE: 30/7/26

IMPLEMENT PACKET SNIFFING USING RAW SOCKETS IN PYTHON

AIM:

To Implement a Packet Sniffer using raw sockets in python.

REQUIREMENTS

- Python 3.x
- Raw Socket Support
- Windows/Linux operating System (Run with administrator or root privileges).

PACKET SNIFFING:

Packet Sniffing is the process of Capturing and analyzing network packets as they travel across a network. In python, raw sockets provide direct access to network packets, allowing programmers to inspect packet headers and payloads. This technique is widely used for network monitoring, troubleshooting and Security analysis.

OUTPUT SAMPLE 1

Packet Captured
Source IP : 192.168.1.10
Destination IP : 192.168.1.1
Protocol : TCP
Packet Length : 60 bytes

OUTPUT SAMPLE 2

Packet Captured
Source IP : 8.8.8.8
Destination IP : 192.168.1.10
Protocol : ICMP
Packet Length : 84 bytes

PROCEDURE:

1. Import the required Python modules
2. Create a raw Socket.
3. Bind the socket to the network interface
4. Continuosly listen for incoming packets.
5. Capture the packet data
6. Display packet information such as Source, destination and protocol
7. Repeat until the program is terminated

PROGRAM

```
import socket
s = socket.socket(socket.AF_PACKET, socket.SOCK_RAW,
                  socket.ntohs(3))
while True:
    pkt = s.recvfrom(65535)[0]
    if len(pkt) > 34 and pkt[12:14] == b'\x08\x00':
        print(f"Packet Captured \n Source IP : {socket.inet_ntoa(pkt[26:30])}
        \nDestination IP : {socket.inet_ntoa(pkt[30:34])}\n Protocol :
        {pkt[23]}\n Packet Length: {len(pkt)}\n")
```

RESULT:

Thus the packet Sniffer using raw Socket in Python implemented Successfully.